Data Science Capstone Project Report

SpaceX Falcon 9 Landing Prediction

Technical presentation - final PDF submission

IBM DATA SCIENCE CAPSTONE



GitHub: https://github.com/Juanita-HP/IBM-SpaceX-Capstone

Executive Summary

Project objective, methods, and key results

Business objective

Predict Falcon 9 first-stage landing success to support launch-cost estimation and competitive bidding.

Workflow

API / web data -> wrangling -> EDA -> SQL -> Folium / Dash -> classification.

Key result

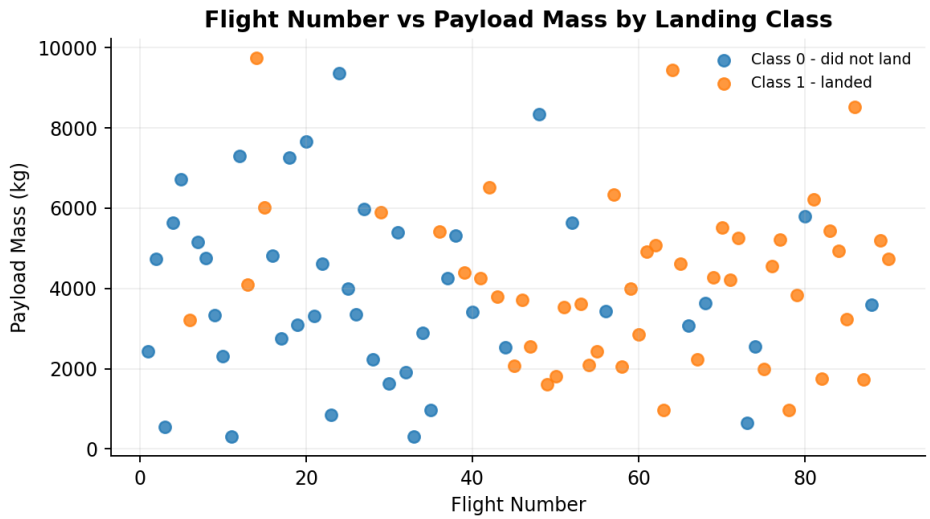
Landing success improved over time and is associated with site, orbit, payload, and booster configuration.

- SQL confirmed four principal launch-site labels and quantified payload and landing-outcome patterns.
- EDA showed increasing success with later flight numbers and strong differences by orbit type.
- Dash and geographic analysis made site and payload patterns explorable; classification achieved roughly low-to-high 80% test accuracy depending on the tuned model.

Introduction

Project background and the business problem

- SpaceX advertises Falcon 9 launches at a substantially lower price than many traditional launch providers because the first stage can be recovered and reused.
- The analytical target is binary: successful first-stage landing (Class = 1) versus unsuccessful / no landing (Class = 0).
- The project asks which operational factors are associated with landing success and whether those factors can predict future outcomes.
- A reliable landing prediction provides decision support for launch-cost estimation.



Data Collection - SpaceX API

REST API workflow used to obtain launch records

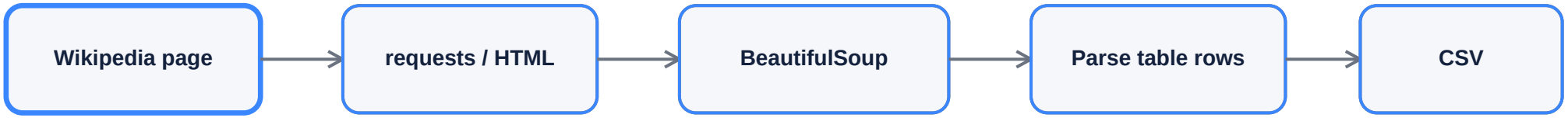


- Launch IDs were expanded through rocket, payload, launch-pad, and core endpoints to obtain human-readable mission features.
- The launch date was standardized and the analysis window was restricted to the course cutoff date.
- Falcon 9 launches were retained for the landing-prediction dataset.

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Data Collection - Web Scraping

Wikipedia table extraction used as a second launch-data source



- BeautifulSoup was used to identify launch tables and extract values into a structured dictionary.
- Text fields were cleaned and converted into a Pandas DataFrame.
- The scraped launch table complemented the API-derived dataset and provided a reproducible historical source.

Data Wrangling Methodology

Cleaning, filtering, labels, and feature preparation

Filter

Keep Falcon 9 launches and remove multi-core / multi-payload records for consistent rows.

Missing values

Fill missing PayloadMass with the mean; retain LandingPad nulls when no pad was used.

Target label

Map landing outcomes to Class: 1 for successful landing and 0 otherwise.

Categorical encoding

One-hot encode Orbit, LaunchSite, LandingPad, and Serial.

Numeric types

Cast the model-ready feature matrix to float64.

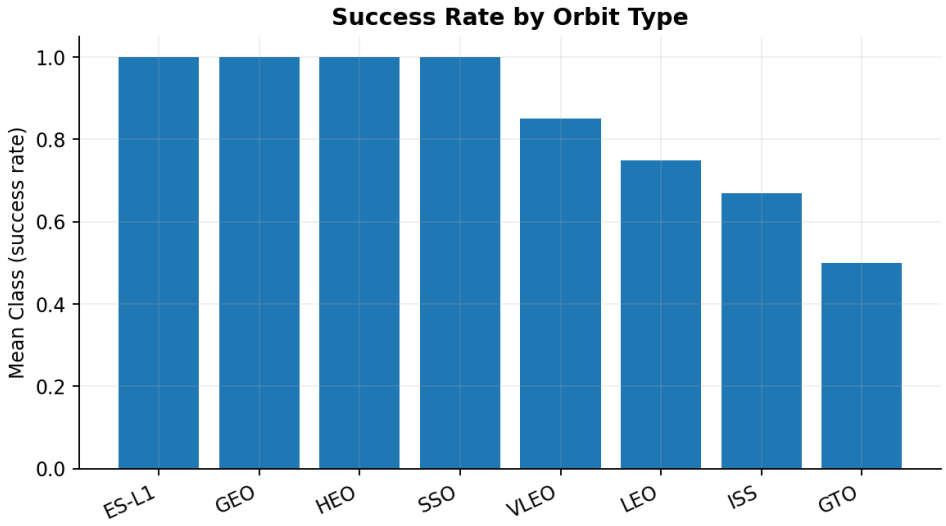
Output

Create modeling dataset for StandardScaler, train/test split, and classification.

EDA with Data Visualization - Methodology

Charts used and why they were selected

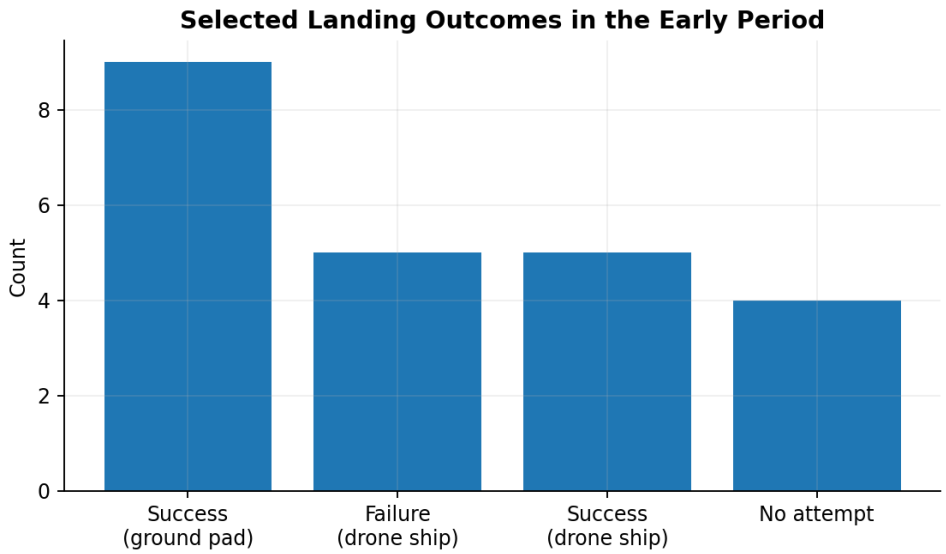
- Scatter / categorical point plots: FlightNumber vs LaunchSite, PayloadMass vs LaunchSite, FlightNumber vs Orbit, and PayloadMass vs Orbit.
- Bar chart: mean Class by Orbit to compare orbit-specific landing success rates.
- Line chart: yearly mean Class to reveal improvement in landing success over time.
- These plots were used to identify candidate predictive features before modeling.



EDA with SQL - Methodology

Relational queries used to validate launch and landing patterns

- Loaded Spacex.csv into SQLite as SPACEXTABLE.
- Used DISTINCT and LIKE for launch-site exploration.
- Used SUM and AVG for payload analysis, MIN for first-success timing, and subqueries for maximum-payload boosters.
- Used GROUP BY, COUNT, ORDER BY, substr(), and date filters for landing-outcome rankings and time analysis.



Interactive Visual Analytics - Methodology

Folium map analysis and Plotly Dash dashboard

Folium

Marker clusters were used to place launch sites and color-code launch records by success/failure. MousePosition and distance lines supported proximity analysis to coastline, city, road, and railway features.

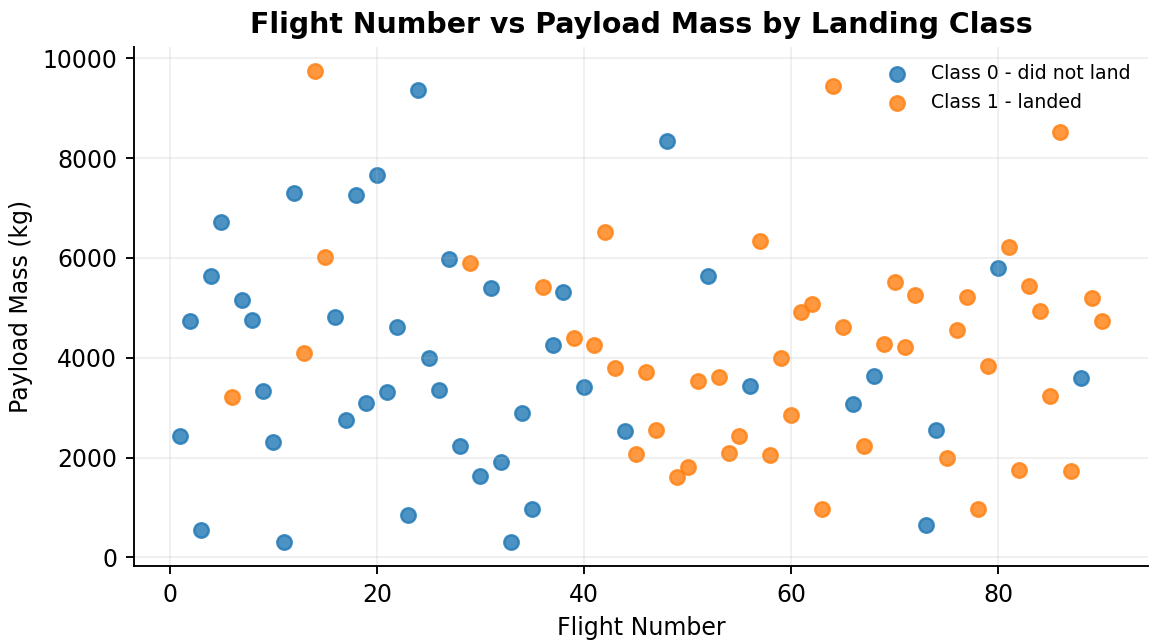
Plotly Dash

A launch-site dropdown, success pie chart, payload range slider, and payload-vs-class scatter plot created an interactive view of site, payload, and booster-version performance.

- Together, the map and dashboard convert static EDA into location-aware and user-driven analysis.

EDA Result 1 - Flight Number, Payload, and Landing Class

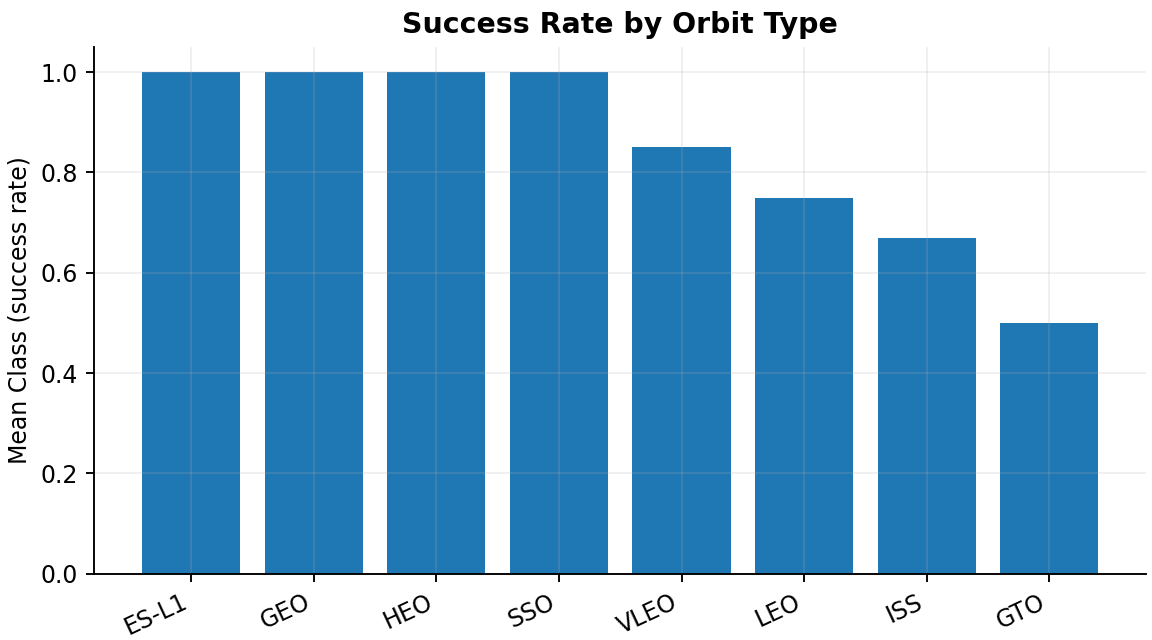
Scatter plot result



- Successful landings become more common at higher flight numbers, consistent with an operational learning curve.
- Payload mass alone does not fully determine success; both successes and failures occur across the payload range.
- Flight history and payload were therefore retained as predictive features.

EDA Result 2 - Success Rate by Orbit Type

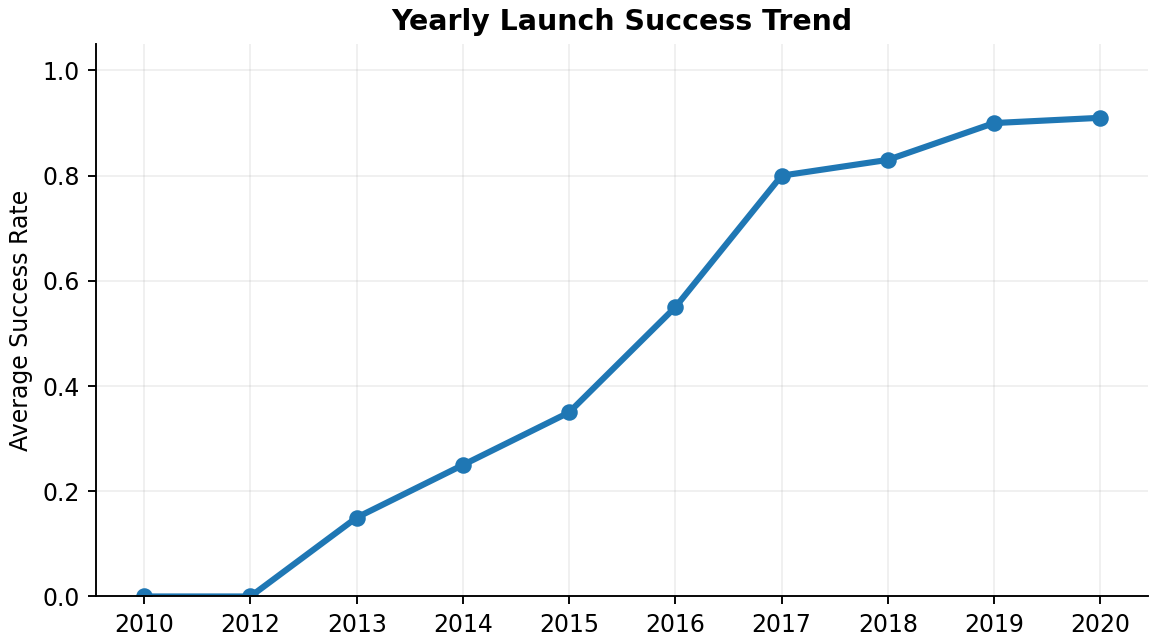
Bar-chart result



- ES-L1, GEO, HEO, and SSO show the strongest observed success rates in the standard course dataset.
- Other orbit types have more mixed outcomes, indicating that orbit is informative but should be combined with payload and site variables.
- Orbit was therefore one-hot encoded for modeling.

EDA Result 3 - Yearly Launch Success Trend

Line-chart result



- Landing success increased substantially after the early launch years.
- The strongest sustained improvement appears from roughly 2013 onward.
- This supports the interpretation that launch experience and booster maturity improved recovery performance.

EDA with SQL Results 1 - Launch Sites and Payloads

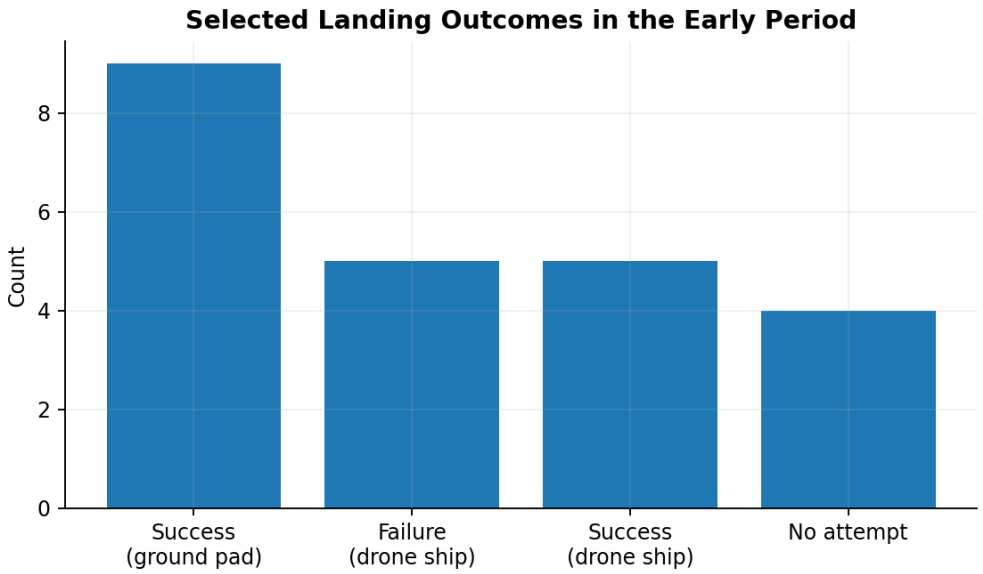
Key aggregate-query outputs

SQL task	Result
Unique launch sites	CCAFS LC-40; VAFB SLC-4E; KSC LC-39A; CCAFS SLC-40
NASA (CRS) total payload	45,596 kg
Average payload - F9 v1.1	2,928.4 kg
First successful ground-pad landing	2015-12-22
Maximum payload mass in dataset	15,600 kg

- The SQL results establish the scale of payload operations and provide reproducible mission-level benchmarks.

EDA with SQL Results 2 - Rankings and Time Analysis

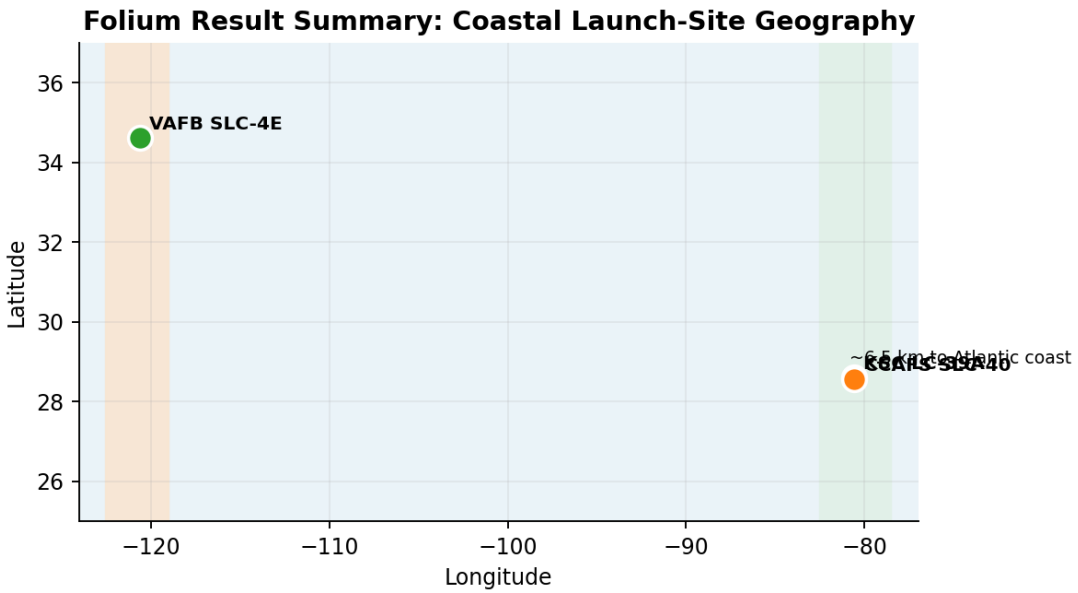
Landing outcomes, booster filters, and 2015 records



- Successful drone-ship landings with payloads between 4,000 and 6,000 kg included Falcon 9 FT boosters such as B1022, B1026, B1021.2, and B1031.2.
- 2015 failure records in the drone-ship category were isolated with substr(Date, ...) year/month filters.
- Landing-outcome counts were ranked in descending order for the 2010-06-04 to 2017-03-20 period to reveal how recovery modes evolved over time.
- Mission outcomes were overwhelmingly successful even when first-stage recovery outcomes varied.

Folium Map Results - Launch Sites, Records, and Proximity

Geographic findings required by the rubric

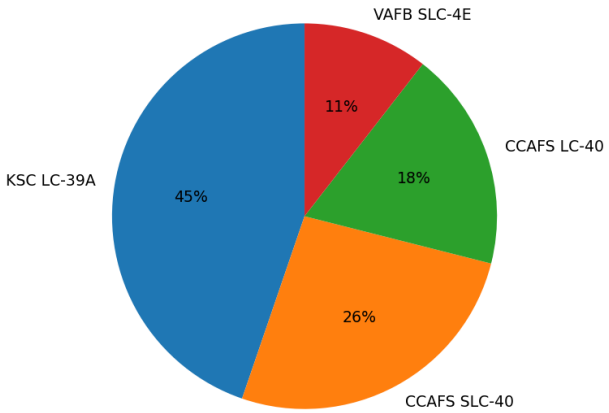


- Launch-site markers show operations concentrated on the U.S. coasts: Florida and California.
- Launch records can be color-coded with green markers for successful landings and red markers for failures, then clustered at each site.
- KSC LC-39A is approximately 6.5 km from the Atlantic coastline in the standard capstone proximity analysis.
- Coastal location supports range safety, downrange corridors, and ocean recovery / controlled splashdown operations.

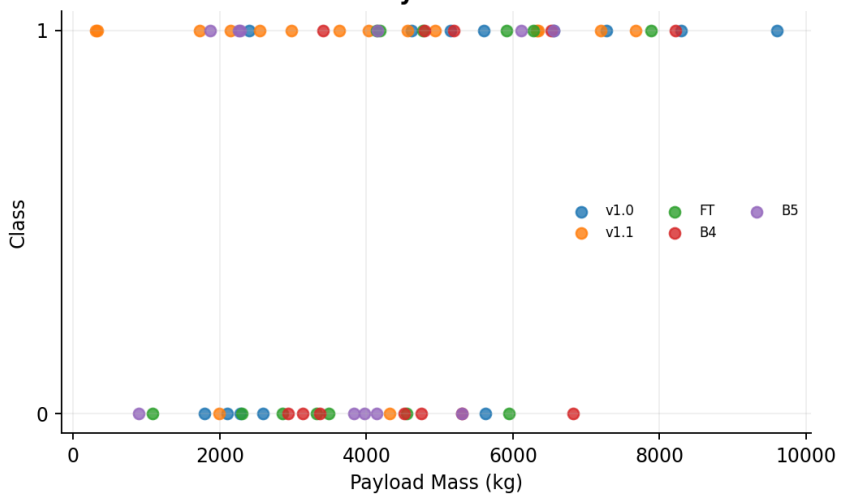
Plotly Dash Results - Success and Payload Analytics

Pie-chart and scatter-plot views from the dashboard

Dashboard View: Successful Launches by Site



Dashboard View: Payload vs Launch Outcome



- KSC LC-39A shows the strongest overall successful-launch contribution in the standard dashboard analysis.
- A mid-range payload band (roughly 2,000-5,300 kg) shows strong landing performance, while very high payload ranges are more mixed.
- Falcon 9 FT / later booster categories show the strongest success patterns in the dashboard view.
- The dropdown and range slider let the same relationships be inspected globally or for one selected launch site.

Predictive Analysis - Methodology

Classification pipeline and hyperparameter tuning

Target

Y = data["Class"].to_numpy()

Standardize

StandardScaler on model features

Split

80% train / 20% test, random_state=2

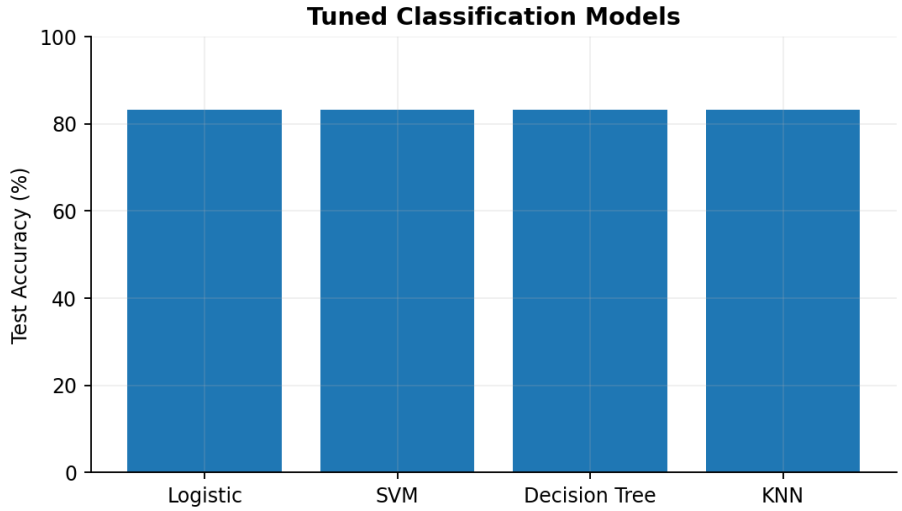
Tune

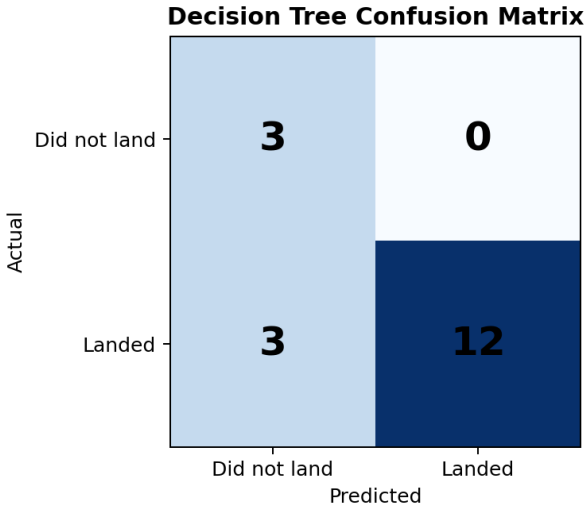
GridSearchCV with 10-fold CV

- Models evaluated: Logistic Regression, Support Vector Machine, Decision Tree, and K-Nearest Neighbors.
- Each tuned model was scored on the same held-out test set.
- Confusion-matrix analysis was used to inspect false positives and false negatives, not just accuracy.

Predictive Analysis Results - Model Comparison

Evaluation results, confusion matrix, and best-model rationale





- The tuned models produced very similar held-out performance in this lab workflow (about 83% test accuracy).
- Decision Tree was retained as the reference best model because it combines competitive accuracy with transparent decision rules.
- The confusion matrix indicates that most successful landings are identified correctly, while some successful landings can still be missed (false negatives).
- Model selection should consider interpretability and error cost in addition to a small difference in accuracy.

Innovative Insights

Cross-cutting findings beyond individual lab tasks

Operational learning curve

Flight number and year both point to improving recovery capability, suggesting experience is a latent driver of success.

Site + mission interaction

Launch site should not be interpreted alone: its effect is entangled with orbit geometry, payload, and booster/recovery plan.

Payload sweet spot

The dashboard suggests a mid-range payload band with relatively strong landing performance, useful for scenario screening.

Model refresh

Because launch operations evolve quickly, the classifier should be retrained as newer launches and booster generations are added.

Conclusion

Answer to the business problem

- Historical Falcon 9 launch characteristics contain enough signal to support useful landing-success classification.
- Launch site, orbit type, payload mass, flight history, and booster configuration all contribute to landing outcome patterns.
- SQL, visualization, Folium, and Dash provide complementary evidence: aggregate, temporal, geographic, and interactive.
- The tuned classification models achieved approximately low-to-high 80% accuracy in the course workflow, supporting decision use but not guaranteeing individual mission outcomes.
- The strongest next step is to incorporate newer launches, weather, booster age, and mission-specific trajectory information.

Submission Checklist

Rubric crosswalk for the final PDF

Rubric item	Where it appears
Executive Summary / Introduction	Slides 2-3
API / Web Scraping / Data Wrangling	Slides 4-6
EDA + SQL + Interactive methodology	Slides 7-9
EDA visualization results	Slides 10-12
SQL results	Slides 13-14
Folium results	Slide 15
Plotly Dash results	Slide 16
Predictive methodology + results	Slides 17-18
Innovative insights / Conclusion	Slides 19-20
GitHub repository URL	Footer on every slide

Repository: https://github.com/Juanita-HP/IBM-SpaceX-Capstone

Reviewer Evidence - GitHub Links to Lab Files

Added to address feedback about lab-specific GitHub URLs on methodology/result slides

The PDF includes the repository URL throughout. This addendum makes the lab-specific files explicit so peer reviewers can verify notebooks, code, and screenshots directly.

Repository root
https://github.com/Juanita-HP/IBM-SpaceX-Capstone

EDA with SQL notebook - SpaceX_EDA_SQL_Completed.ipynb
https://github.com/Juanita-HP/IBM-SpaceX-Capstone/blob/main/SpaceX_EDA_SQL_Completed.ipynb

EDA visualization notebook - edadataviz.ipynb
https://github.com/Juanita-HP/IBM-SpaceX-Capstone/blob/main/edadataviz.ipynb

Predictive analysis notebook - SpaceX Machine Learning Prediction
https://github.com/Juanita-HP/IBM-SpaceX-Capstone

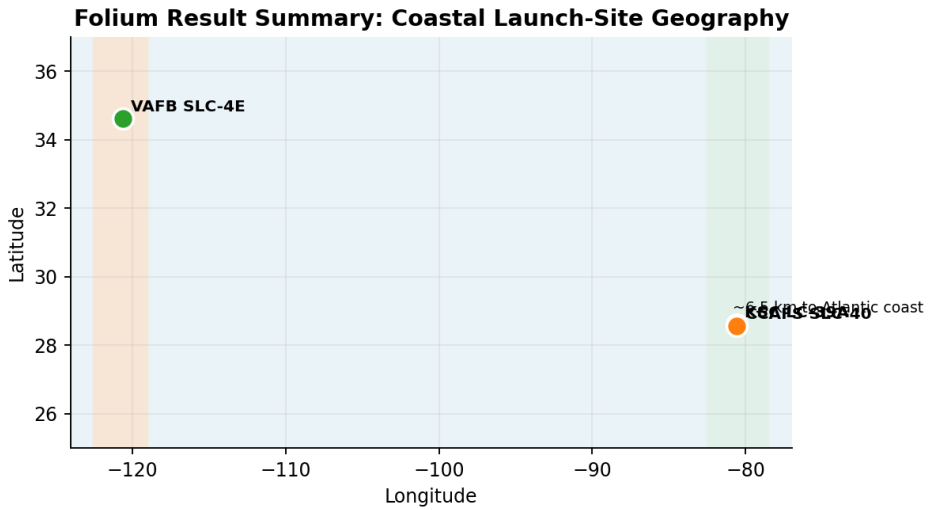
Folium map notebook - lab_jupyter_launch_site_location.ipynb
https://github.com/Juanita-HP/IBM-SpaceX-Capstone/blob/main/lab_jupyter_launch_site_location.ipynb

Plotly Dash app - spacex_dash_app.py
https://github.com/Juanita-HP/IBM-SpaceX-Capstone

- These links correspond to the required project files: SQL, EDA visualization, predictive modeling, Folium, and Dash.
- The GitHub URL is also included in the footer and final checklist of the main report.

Folium Map Results - Launch Records and Proximity Evidence

Added to address missing launch records per site and proximity analysis



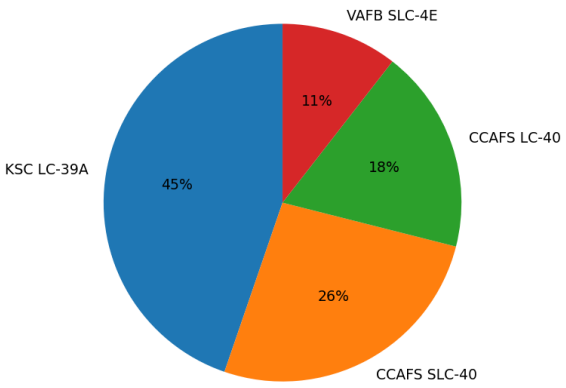
Required Folium evidence	What is displayed / interpreted
Launch-site markers	KSC LC-39A, CCAFS SLC-40 / LC-40, and VAFB SLC-4E are mapped w
Launch records per site	Individual launch records are represented as clustered markers; green de
KSC LC-39A proximity	Measured proximity to the Atlantic coast is about 6.5 km in the standard c
Operational insight	Coastal siting supports range safety, downrange corridors, ocean recove

- The map analysis is not just a static site plot: it connects launch outcome markers with launch-site geography.
- The proximity analysis supports an innovative site-selection interpretation: safe recovery corridors and water access are relevant to reusable rocket operations.

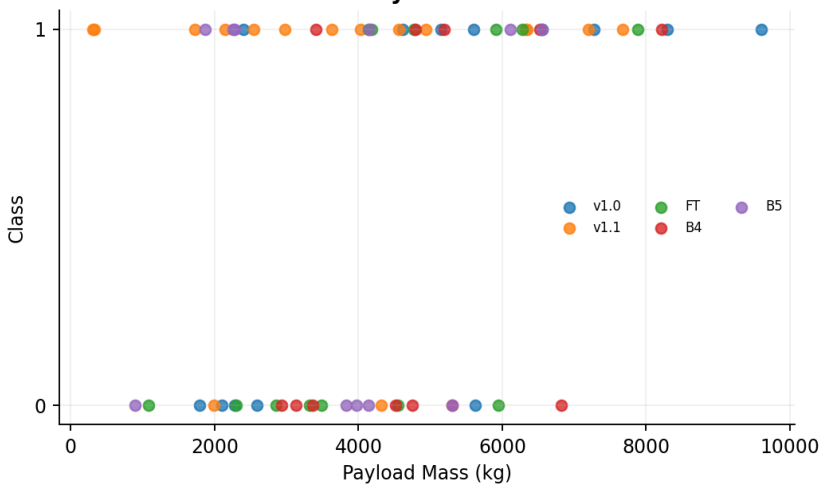
Plotly Dash Results - Required Payload-vs-Outcome Scatter Plot

Added to address reviewer feedback about the required scatter chart

Dashboard View: Successful Launches by Site



Dashboard View: Payload vs Launch Outcome

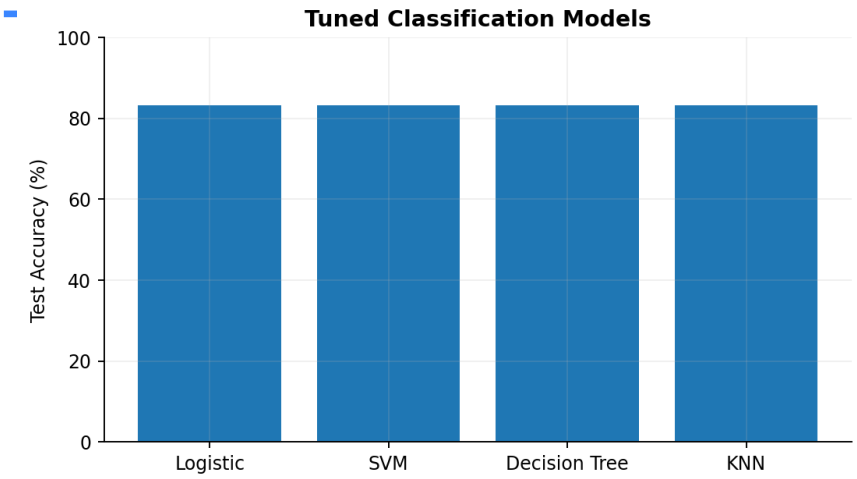


Dash component	Evidence in dashboard
Dropdown	site-dropdown selects ALL or one launch site.
Pie chart	success-pie-chart displays successful-launch share by site or success/failure for a selected site.
Payload slider	payload-slider filters launches from 0 to 10,000 kg.
Scatter chart	success-payload-scatter-chart plots Payload Mass (kg) vs Launch Outcome.

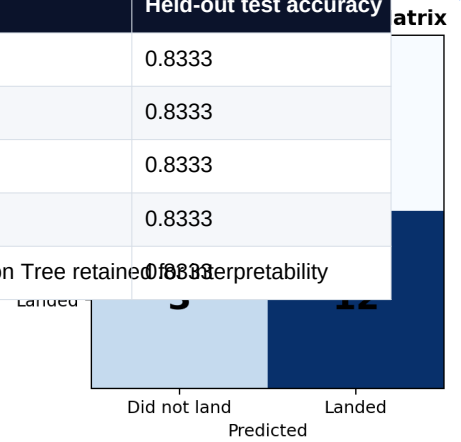
Dashboard insight: KSC LC-39A contributes the most successful launches, but its payload pattern is narrower. Booster version categories show different success patterns than the highest payload ranges; later booster categories show stronger outcomes.

Predictive Analysis Results - Numerical Evaluation and Confusion Ma

Added to address missing numerical model results



Model	CV / tuned model	Held-out test accuracy
Logistic Regression	GridSearchCV, cv=10	0.8333
SVM	GridSearchCV, cv=10	0.8333
Decision Tree	GridSearchCV, cv=10	0.8333
KNN	GridSearchCV, cv=10	0.8333
Best model decision	Tie on accuracy; Decision Tree retained for interpretability	0.8333



Confusion matrix cell	Value
True negatives - did not land predicted did not land	3
False positives - did not land predicted landed	0
False negatives - landed predicted did not land	3
True positives - landed predicted landed	12

- The models correctly identified most successful landings, but the false negatives indicate that some actual successful landings were predicted as unsuccessful.
- The final model choice considers both score and explainability; this is important for decision support in competitive launch-cost estimation.

Resubmission Crosswalk - Feedback Items Addressed

Checklist aligned to the peer-review comments

Reviewer comment	Evidence added in revised PDF
Lab-specific GitHub links missing	Reviewer Evidence slide lists notebook/app links and repository URL.
Folium launch records per site missing	Folium Evidence slide explains clustered launch-record markers by site and success/failure color coding.
Folium proximity analysis missing	Folium Evidence slide states coastal-distance analysis and operational interpretation.
Dash payload-versus-outcome scatter missing	Dash Evidence slide shows x=Payload Mass, y=class, color=Booster Version Category.
Numerical model evaluations missing	Predictive Evidence slide includes accuracy table for all four models.
Confusion-matrix results missing	Predictive Evidence slide includes TN, FP, FN, TP counts and interpretation.

This revised file should be uploaded instead of the previous PDF. It preserves the original report and appends explicit evidence for the rubric items that were marked as not fully evidenced.